HOW SERGE WORKS

Logic & decision trees of the Serge coding agent — the free-tier hive with one personality. Snapshot: 2026-07-11. Runs on Linux, macOS (launchd), and Windows (WSL2 or Git Bash).

1 What Serge is

Serge is a terminal AI coding agent — a fork of *openclaude* (a Claude-Code-compatible engine) that runs entirely on **cloud free tiers** (Mistral, Google, Cerebras), wired together by a local **LiteLLM router** on `localhost:4000`. It behaves like a small **hive** of specialists: different models take different jobs, but every seat is injected with the same **constitution** (`~/.serge/CONSTITUTION.trimmed.v3.md`) — one personality, many brains. Around the engine sits a shell of **hooks**: session-start memory loaders, an end-of-turn verify→review→gate pipeline, a spend watchdog, and autonomous loops that run Serge headlessly on timers.

> **THE GOLDEN RULE:** plentiful seats by default; the scarce deep seat only when it earns it. The cheapest brain that CAN do the job — and nothing ships unchecked.

2 The hive — seats & roster

Seat name	Model (2026-07)	~Free quota	Role
`local-coder`	Mistral Large	**~1B tok/month**	**Default workhorse** — ~90% of turns
`fast-coder`	Devstral Medium	**(same Mistral pool)**	The scout — codebase recon, burst fan-outs
`think-coder`	Magistral Medium	**(same Mistral pool)**	Deep reasoner — adversarial passes, /deepthink
`qwen-coder`	gpt-oss-120b (Cerebras)	**~1M tok/day**	**The reviewer** — independent second opinion, different vendor on purpose
`pro-coder`	Gemini 3.1 Flash-Lite	**own daily bucket**	Classifier / live-fact single-shot calls (kept off the driver's quota)
`cloud-brain` / `bedrock-brain`	Gemini 3.5 Flash	**~20 req/day**	**The architect** — hardest reasoning + the only VISION seat
`free-flash / -flash3 / -flash25`	Gemini buckets	**~20/day each**	Fallback rungs — per-model quota, so each hop is fresh budget
`free-qwen`	Qwen3 Coder 480B (OpenRouter :free)	**~50 req/day**	4th-provider rung before anything paid
`opus-paid / haiku-paid / kimi-coder`	Claude Opus 4.6 / Haiku 4.5 / Kimi K2.5 (Bedrock)	**PAID**	Escape hatches — bill only when named; opus is in NO fallback chain

Seats are defined in `~/.serge/litellm.yaml`; the router hides them all behind one OpenAI-compatible endpoint, so the engine just asks for a model name.

3 Decision tree — launching `serge`

```
$ serge [--cloud|--council|--terse|--auto|--yolo|--uncap]
   |
   |-- source ~/.serge/serge.env          (search keys, routing knobs, budgets)
   |-- parse mode flags
   |     --council -> append council.md system prompt, SERGE_REVIEW_ENFORCE=1
   |     --terse   -> append lean.md output-discipline overlay
   |     --auto    -> permission-mode acceptEdits   (auto-apply edits only)
   |     --yolo    -> bypassPermissions + ENFORCE review hooks as compensation
   |
   |-- daily-spend cap check (~/.serge/monitor/.budget-capped)
   |     capped?  --uncap given -> clear flag, re-baseline today's spend, go
   |              else          -> explain the pause + how to resume, EXIT
   |
   |-- router health: GET localhost:4000/v1/models
   |     down? -> start it, wait <=45s (LiteLLM cold boot ~30s):
   |               Linux: systemctl --user start serge-router
   |               no systemd (macOS / Git Bash / bare WSL): spawn litellm
   |                 directly with router.env loaded -> monitor/router.log
   |     still down? -> print fix commands, EXIT
   |
   '-- exec node dist/cli.mjs   with OPENAI_MODEL = local-coder   (default)
                                 or bedrock-brain (--cloud)
                                 [OPENAI_MODEL=<seat> overrides per session]
```

4 Decision tree — which brain answers this turn?

Re-evaluated on **every human turn** inside the engine's OpenAI shim (`openaiShim.ts`); injected `<system-reminder>` text is stripped before classifying so hidden context can't trigger escalation. Escalations hold for the whole tool loop, then flip back on the next easy turn.

```
 You send a prompt
   |
   v
 [1] session started with --cloud? .....................> bedrock-brain (Gemini 3.5)
   |
 [2] message contains an IMAGE? ........................> SERGE_VISION_MODEL
   |      (only the brain seat has vision)                (= bedrock-brain)
   |
 [3] live-fact question?  (scores, prices, weather,
   |   news — cheap coders hallucinate current facts) ..> SERGE_LIVEFACT_MODEL
   |                                                       (= cloud-brain)
 [4] "hard task" classifier regex hits?
   |   architecture / redesign / trade-offs / race
   |   condition / deadlock / memory leak / root-cause /
   |   security-critical / crypto / distributed txn ...
   |   PLUS practical-debug signals: "still not working",
   |   "error when", "keeps crashing", stack trace,
   |   regression, build failure .......................> SERGE_CLASSIFIER_MODEL
   |                                                       (= cloud-brain)
 [5] otherwise — the usual ~90% .......................> local-coder (Mistral) [DEFAULT]

 Reasoning depth (orthogonal): /effort low..xhigh
   - seats in SERGE_THINKING_SEATS (opus/haiku-paid only) get native Anthropic
     thinking budgets (xhigh = 16384 tokens)
   - every other seat gets reasoning_effort (xhigh clamps to high);
     Gemini/gpt-oss/Magistral honor it, Mistral Large ignores it (non-reasoner)
```

5 Fallback chains — when a seat 429s or dies mid-stream

Fallbacks fire **only on error** (throttle / 5xx / mid-stream drop). Chains hop **providers** — one account's outage or quota can never collapse a whole chain. Gemini free quota is per-model, so each Gemini bucket is a fresh ~20/day. Only the workhorse chain ends on a paid rung (Serge must not brick mid-task); everything else stays free and fails loudly.

```
local-coder (Mistral Large)                                    THE WORKHORSE CHAIN
  -> free-flash25 (Gemini 2.5 Flash) -> free-flash3 (Gemini 3 Flash)
  -> free-flash (3.1 Flash-Lite)     -> free-qwen (OpenRouter :free)
  -> haiku-paid (Bedrock, PAID — the only paid rung anywhere)

cloud-brain / bedrock-brain (Gemini 3.5)                       THE BRAIN CHAIN
  -> free-flash3 -> free-flash25 -> free-flash -> local-coder
     (Gemini buckets first — they drive agentic loops fine; NEVER Cerebras,
      whose models hang Claude-Code-style tool loops — eval-verified)

qwen-coder (reviewer) -> pro-coder -> free-flash        opus-paid -> (nothing:
fast-coder (scout)    -> free-flash -> free-scout        fails loudly rather
think-coder (reasoner)-> free-brain -> pro-coder         than bill silently)
```

6 Decision tree — end of turn (the Stop pipeline)

Two Stop hooks run in order (`settings.json`). Exit code 2 from a hook **blocks the stop** and feeds its message back to Serge — that's how the hive forces itself to fix things.

```
Serge tries to end its turn
  |
  v
[A] continue-on-unfinished.sh          THE ANTI-STALL CHECK
  |   transcript claims ongoing work ("let me check...") but no tool ran,
  |   or claims "done" with zero tools used?
  |      YES -> exit 2: "finish the job" (anti-loop cap; disable:
  |             SERGE_AUTOCONTINUE_DISABLE)          -> Serge keeps working
  |      no  -> continue
  v
[B] stop-checks.sh  — one process, three stages:
  |
  |  STAGE 1: VERIFY (free, deterministic — verify-on-stop.sh)
  |    typecheck + lint + hygiene greps (debug lines, merge markers,
  |    .only tests) on files edited this turn
  |      FAIL + enforce -> exit 2, Serge fixes, re-runs ONCE
  |                        (repeat failures land in the reflexion log)
  |      hard fail      -> SKIP the review (code is about to change)
  |
  |  STAGE 2: REVIEW (free AI — review-on-stop.sh)
  |    qwen-coder (gpt-oss, DIFFERENT VENDOR) reads the turn's diff
  |      BLOCKING bug + enforce (--council/--yolo) -> brain seat writes a
  |                       fix plan -> exit 2 -> Serge applies it, ONCE
  |      nits / clean   -> advisory note only
  |
  |  STAGE 3: GATE (only if Serge's OWN brain files changed — gate-on-stop.sh)
  |    files in SERGE_GATE_HASH_GLOBS (constitution, council, agents) hashed;
  |    changed? -> run the golden-task eval suite vs baseline
  |      regression -> exit 2: revert-or-justify   (the self-edit ratchet;
  |                    a PreToolUse hook also gates constitution WRITES)
  v
[C] turn ends -> reflexion-log.sh appends lessons; statusline shows session cost
```

7 Session start — what Serge remembers

Loader (SessionStart hook)	Injects
`memory-load.sh`	The memory INDEX (`memory/MEMORY.md`) — one line per durable fact; full fact files opened only when relevant. Add facts with `/remember`.
`progress-load.sh`	Where it left off on ongoing tasks (also after /clear & compaction).
`reflexion-load.sh`	Past REPEATED verify failures — real tool output, so it stops tripping on the same stone.
`plan-pointer.sh`	Pointer to a previously approved plan (`plan.md`) — plan-high / execute-cheap.
`notifications-load.sh`	Pending items from the autonomous loops (`NOTIFICATIONS.md`).
`stale-binary-check.sh`	Warns when the running build is older than the source (a /clear keeps old code!).

8 Autonomous loops — Serge without prompts

Outer loops (`~/.serge/loops/`) prompt Serge **headlessly** on triggers; the human feeds queues and reviews results. Every loop follows the same shape:

```
trigger (timer / path-watch) -> guards (kill switches, daily caps, locks)
  -> build_prompt()  = the SENSOR (deterministic; often decides "nothing to do" for $0)
  -> serge -p --yolo = the ACTUATOR (bounded: --max-turns, budget, timeout(1))
  -> tripwire        (did Serge touch its OWN brain files? -> ALL loops hard-disable)
  -> journal (one JSON line/run) -> post_run() (branches, inboxes, notifications)
```

Loop	Trigger	Sensor -> action
`eval-gate`	nightly 03:30	Run the golden-task gate; on regression: diagnose, write findings, **propose** (never apply) constitution fixes.
`err-triage`	`serge-query-errors.log` grows	Classify new errors vs known causes → `triage-inbox.md` ; new signatures get a draft memory note for human promotion.
`backlog`	hourly (ships **disabled**)	Drain one `- [ ]` item from an enrolled project's BACKLOG.md in an isolated worktree on branch `loop/<slug>` ; tests gate the ✔.

Rails: global kill switch `touch ~/.serge/loops/DISABLED` (per-loop: `<name>/DISABLED`); per-loop + global daily run caps; every run bounded in turns, dollars and wall-clock; loops may only ever **propose** changes to the brain — the tripwire + eval ratchet enforce it.

9 Memory & self-improvement

```
                    +-------------------------------------+
task completed ---->| reflexion-log.sh: what failed, what   |----+
verify failed ----->| worked (real tool output, not vibes)  |    |
                    +-------------------------------------+    v
/remember  --> ~/.serge/memory/.md  (index: MEMORY.md, loaded each session)
                                                               |
bigger lesson? ----> constitution-proposals.md   (HUMAN reviews & applies)
                           |
                           v
     gate-on-constitution-edit.sh (PreToolUse) + nightly eval-gate loop:
     any brain edit must NOT regress the 10 golden tasks vs baseline
     (01 smoke ... 05 no-hardcoded-secret ... 07 honesty>approval ... 10 no-fabricated-data)
```

Net effect: Serge learns durable facts freely, but its **personality can only be changed through a regression-gated, human-approved path** — even by itself.

10 Cost discipline

Mechanism	What it does
All-free roster	Every working seat is a $0 tier; session cost line normally reads ≈ $0.00.
Budget watchdog (5-min timer)	Pauses the router when paid (OpenRouter) daily spend crosses the cap (drop-in: $10/day). `serge --uncap` or UTC midnight resumes.
Paid-seat tripwire	Any Bedrock escape-hatch use is flagged in `NOTIFICATIONS.md` — a paid seat can never fire silently.
Quota gauge	`quota.sh` feeds the statusline: today's brain-seat usage vs the ~20/day bucket.
Free-tier scanner (weekly)	Probes providers for new/removed free models; proposes roster upgrades.
Thinking tripwire (daily)	Probes that reasoning depth still actually fires on key seats (providers silently change APIs).

11 Quick reference

Modes & flags

```
serge               the free workhorse
serge --cloud       start on the brain seat
serge --council     architect plans, reviewer attacks
serge --terse       lean answers (style only)
serge --auto        auto-apply edits
serge --yolo        hands-off; review hooks enforced
serge --uncap       clear the daily-spend pause
OPENAI_MODEL=<seat> serge    pin any seat
/effort xhigh       max reasoning depth
/cost /recap /hive /remember /deepthink
```

Where things live

```
~/.serge/CONSTITUTION.trimmed.v3.md  the personality
~/.serge/litellm.yaml                seats + fallbacks
~/.serge/settings.json               hook wiring
~/.serge/stop-checks.sh              verify->review->gate
~/.serge/loops/                      autonomous loops
~/.serge/memory/                     what Serge remembers
~/.serge/router.env serge.env        THE ONLY KEY FILES
~/programs/serge-0.1.0/              engine (fork) + wrapper
```

In one line: a free workhorse does the work, a second vendor checks it for free, the scarce deep brain is summoned only for the genuinely hard 10% — and the whole hive runs on $0.

Generated 2026-07-11 from the live install (roster of 2026-07-10). Source: `docs/how-serge-works.html` — edit and re-render with the command in the file header. Companion docs: `serge-0.1.0/SERGE.md` (full reference), `serge-0.1.0/HOW_SERGE_WORKS.txt` (plain-text map), `dot-serge/loops/README.md` (loops).